

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-  
a0fd652d46ee96615.jsonl`  
- SHA-256: `f92692a60ccc83c3d3eef511816133cd52c6417a0df1076319e38608cd32b783`  
- Source modified: `2026-06-06T17:31:57+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:31:13.033Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** <https://hackmd.io/@TUIK/rJkRW54cU>

****Title:**** Shuttlecock Trajectory Dataset (TrackNet / TrackNetV2 training data) ? official HackMD release page

****Found via:**** datasets/licensing search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:31:14.883Z)

I'll fetch the source page and extract the relevant claims.

3. user (2026-06-06T17:31:29.876Z)

Shuttlecock Trajectory Dataset Summary

Dataset Contents

The dataset comprises ****26 broadcast badminton videos**** (1280×720 resolution, 30 fps) totaling ****78,200 frames****:

- 23 professional match videos (68,675 frames)
- 3 amateur game videos (9,525 frames)

****Annotations include:**** Shuttlecock trajectory data with X/Y pixel coordinates, visibility flags (0=invisible, 1=visible), and frame numbers for each rally.

License & Commercial Use Terms

****No license information is provided**** in the document. No explicit terms of service, usage restrictions, or commercial use permissions are stated.

Download Source

Dataset available at: ``https://nycu1-my.sharepoint.com/:u:/g/personal/tik_m365_nycu_edu_tw/EWisYhAiai9Ju7L-tQp0ykeBZJd9VQkKqsFrjcqYIDP-g?e=S0AB1Z`` (SharePoint link)

Institution & Contributors

****Contributors:**** `???????Tsì-Uí ?k` (affiliated with NYCU based on SharePoint domain)

Publish Date

Not specified in the document.

Usage Restrictions

****None explicitly stated.**** The dataset is "created for the model training and testing of TrackNet and TrackNetV2," but no formal restrictions on use cases are documented.

****Note:**** Consult the institution directly regarding licensing and usage permissions, as these details are absent from the provided documentation.

4. assistant (2026-06-06T17:31:43.923Z)

The fetch succeeded. Let me capture the key details for the structured output.

The page is the official HackMD release page for the Shuttlecock Trajectory Dataset (TrackNet/TrackNetV2 training data) from NYCU (National Yang Ming Chiao Tung University). The most research-relevant finding is the dataset's contents and ? critically for the commercial-licensing question ? the absence of any explicit license.

5. user (2026-06-06T17:31:50.863Z)

Structured output provided successfully

6. assistant (2026-06-06T17:31:57.617Z)

Extracted and returned the structured output. Summary of what I found at <https://hackmd.io/@TUIK/rJkRW54cU>:

- Source quality: primary (the official NYCU/TrackNet authors' own HackMD release page).
- Most research-relevant finding: the dataset provides exactly the per-frame (x, y, visibility) shuttlecock signal the project's WASB substrate already produces, but the release page carries no license and no commercial-use grant ? a hard blocker for the project's permissive/commercial-licensing constraint without contacting the institution.
- Supporting facts captured: 26 badminton videos / 78,200 frames (23 pro + 3 amateur), released by NYCU-affiliated contributors as TrackNet/TrackNetV2 training data, distributed via a personal SharePoint link.
- No publish date was available on the page.